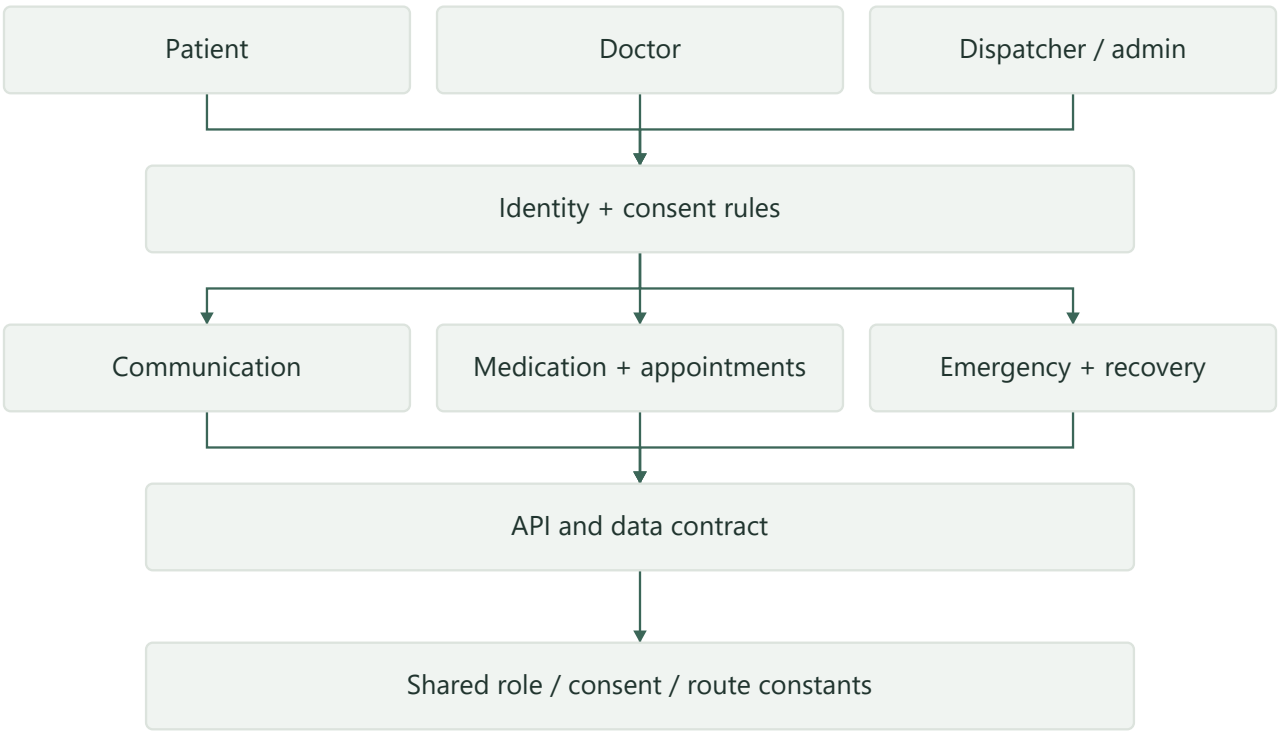
LifeRoute

Healthcare coordination system design

A collaborative system specification for patient communication, medication, appointments, emergency coordination, and recovery tracking.

Architecture



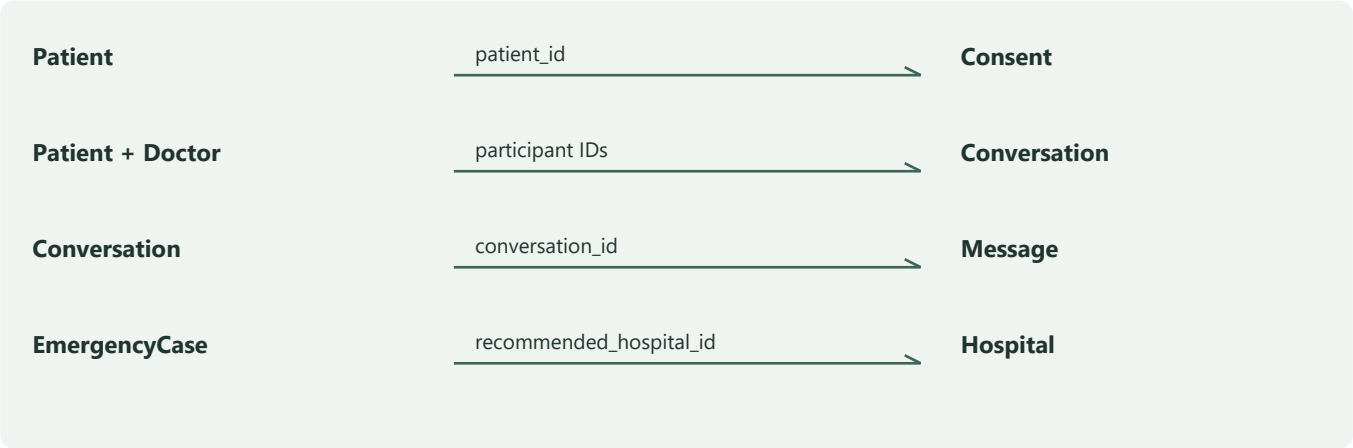
Component overview derived from the supplied source; version-specific details may differ.

Component	Implementation	Role
Roles	Patient, doctor, dispatcher, admin	Access boundaries
Consent	Defined consent modes	Sharing choices
Workflows	Communication and treatment coordination	Specified features
Contract	Entities, API prefix, permissions	Implementation agreement
Shared constants	JSON definitions	Consistent identifiers

Design-stage project

Data model & ownership

LifeRoute is a software system specification. Its contract separates identity, communication, treatment coordination, emergency routing and recovery records. The available material defines an implementation agreement and shared constants; it does not establish a deployed healthcare service.



Identity and relationships

User IDs connect patient and doctor profiles to conversations, prescriptions and appointments. Messages belong to conversations and identify a sender. Consent belongs to a patient and can list permitted doctor IDs. Emergency cases may contain a patient ID, dispatcher assignment and hospital recommendation, while hospitals carry location, vacancy, price and quality fields. IDs are strings and timestamps are intended to use ISO-8601.

Cross-module contract

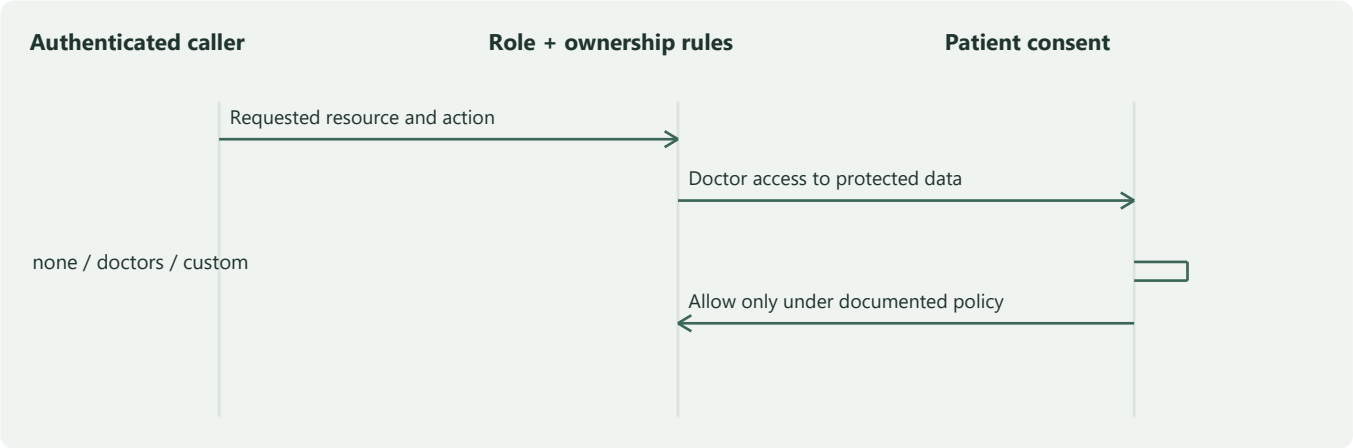
The common /api/v1 prefix and role/consent constants let separately owned modules agree on identifiers and route conventions. The contract assigns Arham identity, consent, emergency management, online consultation and reminders; Dev communication, prescriptions and appointments; and Dau recovery tracking. These are specified responsibilities, not evidence that each implementation is complete.

Entity group	Design purpose
Conversation -> Message	Associate communication with patient/doctor participants
Patient -> Consent	Express the selected medical-data access mode
EmergencyCase -> Hospital	Attach recommendations and dispatcher state to a case

Source / interpretation: LifeRoute/docs/contract_v1.md; shared/constants/api.json, roles.json and consent_modes.json. Relationship diagram is derived from specified identifier fields; no database foreign keys are asserted.

Consent and API behavior

The contract distinguishes authentication, role permission and patient consent. Authentication identifies a caller; role rules decide which features the caller can access; consent adds a patient-specific condition to prescriptions and recovery data. These layers need distinct implementation checks.



Three consent modes

none denies doctor access to patient medical data; doctors permits any doctor under the stated contract; custom restricts access to listed doctor IDs. Consent is specifically named for prescriptions and recovery logs. The design does not imply that every other data type uses the same policy. Server-side authorization must derive identity from the authenticated request rather than accept a patient ID as proof of access.

Routes as agreements

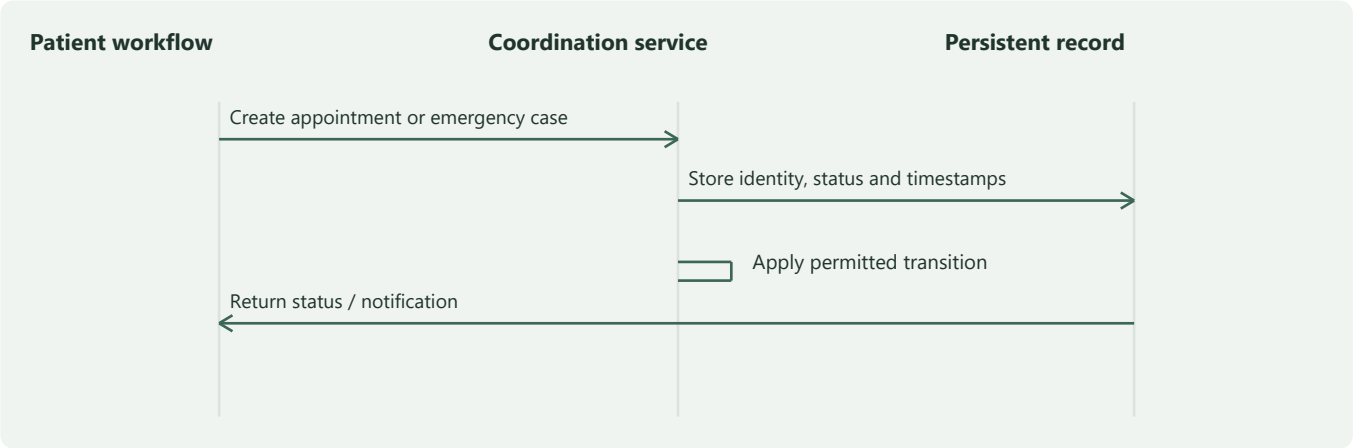
The document specifies JSON responses, bearer authentication except registration/login, a structured error object, and routes for profiles, consent, conversations, prescriptions, appointments, reminders, emergencies and recovery. These are API contracts. Example confidence values and recommendation scores in the document are placeholders, not tested algorithms or clinical findings.

Mode	Doctor access rule
none	Denied for consent-protected medical records
doctors	Any doctor under the specified role policy
custom	Only IDs in allowed_doctor_ids

Source / interpretation: LifeRoute contract sections 2, 5, 6 and permission matrix. This page explains software authorization requirements, not medical advice or a compliance certification.

Workflow boundaries & next implementation

Coordinating several features is more demanding than implementing isolated screens. Appointments, prescriptions, reminders and emergency cases all need coherent state transitions and ownership rules. The contract identifies the entities and endpoints, but many execution details still require design and implementation.



State and concurrency questions

Appointment confirmation should prevent conflicting bookings. Emergency recommendations depend on sufficiently fresh vacancy and location data; the current contract names those fields but does not define a verified ranking algorithm. Reminders include schedule times and a timezone, which requires explicit handling of duplicate delivery and clock changes. These are implementation considerations derived from the design, not existing guarantees.

Incremental verification

A useful first software slice would implement identity, patient profile and consent with tests across all three modes before connecting protected prescriptions or recovery endpoints. Subsequent slices can add communication and scheduling while preserving contributor ownership. Emergency and consultation suggestions should remain bounded by their documented purpose; no medical accuracy or clinical effect is inferred from the specification.

Proposed check	Behavior to establish
Cross-patient request	A caller cannot select another patientâ€™s record merely by changing an ID
Consent revocation	Previously allowed doctor access is re-evaluated
Repeated request / notification	State changes and deliveries have explicit retry behavior

Source / interpretation: LifeRoute contract entities, API endpoints and shared constants. Proposed implementation checks are not claimed as completed tests.

Evidence & source

Implementation, supporting records, and the boundary of the claim

Area	Evidence	Reference
Available artifact	Contract and shared constants	Nine non-history local files
Collaboration	Responsibilities assigned across Arham, Dev and Dau	Contract work-division section
Implementation status	No application implementation established in this copy	Design-stage label retained

Selected implementation files

File / record	Path / reference
contract_v1.md	LifeRoute/docs/contract_v1.md
api.json	LifeRoute/shared/constants/api.json
consent_modes.json	LifeRoute/shared/constants/consent_modes.json
roles.json	LifeRoute/shared/constants/roles.json

The design contract and shared definitions are the available source artifacts.

Evidence boundary

This is a system-design entry. Specified features are not presented as completed medical capabilities, and no clinical outcome is claimed.

Technical note

The contract defines an /api/v1 prefix, string identifiers, ISO timestamps, user roles, consent modes, and coordinated modules. Named contributor responsibilities must remain attached to any implementation narrative.